

Answer to Econ 646 Exam I (2011)*

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March 12, 2012

(A) FIRST ORDER CONDITIONS.

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \{ \log C_t + \theta \log(1 - N_t) + \lambda_t [A_t K_{t-1}^{\alpha} (X_t N_t)^{1-\alpha} + (1 - \delta) K_{t-1} - C_t - K_t] \} \quad (1)$$

F.O.Cs are

$$\frac{\partial \mathcal{L}}{\partial C_t} = \beta^t \left(\frac{1}{C_t} - \lambda_t \right) = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial K_t} = \beta^t (-\lambda_t) + \beta^{t+1} E_t [\lambda_{t+1} (\alpha A_{t+1} K_t^{\alpha-1} (X_{t+1} N_{t+1})^{1-\alpha} + 1 - \delta)] = 0 \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = \beta^t \left(\frac{-\theta}{1 - N_t} + (1 - \alpha) \lambda_t A_t K_{t-1}^{\alpha} (X_t N_t)^{-\alpha} X_t \right) = 0 \quad (4)$$

- The first F.O.C. means that the marginal utility of consumption should equal shadow price of wealth.
- The second F.O.C. shows that the marginal utility brought by increase saving with return in next period, should be the same as shadow price now, or today's marginal utility.
- The last one denotes the marginal utility of labor supply must be equal to marginal utility measure of benefit from wages.

(B) SYSTEM OF EQUATIONS.

From the above analysis, we have the economy described as

$$\begin{cases} \frac{1}{C_t} = \beta E_t \frac{1}{C_{t+1}} [\alpha A_{t+1} K_t^{\alpha-1} (X_{t+1} N_{t+1})^{1-\alpha} + 1 - \delta] \\ \frac{\theta}{1 - N_t} = \frac{1}{C_t} (1 - \alpha) A_t K_{t-1}^{\alpha} (X_t N_t)^{-\alpha} X_t \\ K_t = Y_t + (1 - \delta) K_{t-1} - C_t \\ Y_t = A_t K_{t-1}^{\alpha} (N_t X_t)^{1-\alpha} \end{cases} \quad (5)$$

(C) DE-TRENDING.

*I just show the main points here. You can add something that is important for your argument.

Let $\bar{Y}_t \equiv \frac{Y_t}{X_t}, \bar{K}_t \equiv \frac{K_t}{X_t}, \bar{C}_t \equiv \frac{C_t}{X_t}$. Then, we have

$$Y_t = \bar{Y}_t X_t, K_t = \bar{K}_t X_t, C_t = \bar{C}_t X_t$$

and $X_{t+1} = \gamma X_t$

Thus, for the first equation in (5), we have

$$\frac{1}{\bar{C}_t X_t} = \beta E_t \frac{1}{\bar{C}_{t+1} X_{t+1}} [\alpha A_{t+1} (\bar{K}_t X_t)^{\alpha-1} (X_{t+1} N_{t+1})^{1-\alpha} + 1 - \delta]$$

or

$$\frac{\gamma}{\bar{C}_t} = \beta E_t \frac{1}{\bar{C}_{t+1}} (\alpha \gamma^{1-\alpha} A_{t+1} \bar{K}_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \quad (6)$$

For the second equation in (5), we have

$$\frac{\theta}{1 - N_t} = (1 - \alpha) \frac{1}{\bar{C}_t X_t} A_t (\bar{K}_{t-1} X_{t-1})^\alpha (X_t N_t)^{-\alpha} X_t$$

or

$$\frac{\theta}{1 - N_t} = (1 - \alpha) \frac{1}{\bar{C}_t} \gamma^{-\alpha} A_t \bar{K}_{t-1}^\alpha N_t^{-\alpha} \quad (7)$$

Similarly, for the third equation in (5), we get

$$\bar{K}_t X_t = \bar{Y}_t X_t + (1 - \delta) \bar{K}_{t-1} X_{t-1} - \bar{C}_t X_t$$

or

$$\bar{K}_t = \bar{Y}_t + (1 - \delta) \bar{K}_{t-1} \frac{1}{\gamma} - \bar{C}_t \quad (8)$$

And, for the last equation in (5), the detrended result is

$$\bar{Y}_t = A_t \gamma^{-\alpha} \bar{K}_{t-1}^\alpha N_t^{1-\alpha} \quad (9)$$

(D) INTUITION ON N_t .

At the steady state, we have¹

$$\begin{cases} \frac{\gamma}{\bar{C}} &= \beta \frac{1}{\bar{C}} [\alpha \bar{K}^{\alpha-1} N^{1-\alpha} + 1 - \delta] \\ \frac{\theta}{1-N} &= \frac{1}{\bar{C}} (1 - \alpha) \bar{K}^\alpha N^{-\alpha} \\ \bar{K} &= \bar{K}^\alpha N^{1-\alpha} \gamma^{-\alpha} + (1 - \delta) \bar{K} - \bar{C} \end{cases}$$

$\implies$

¹In this part, I get the solution step by step to show you the details about how to efficiently solve a complicated question like this.

$$\begin{cases} \bar{K}^{\alpha-1} N^{1-\alpha} &= \frac{1}{\alpha} \left(\frac{\gamma}{\beta} + \delta - 1 \right) \equiv D \\ \frac{\theta}{1-N} &= \frac{\bar{K}}{\bar{C}} (1-\alpha) \bar{K}^{\alpha-1} N^{-\alpha} \\ 1 &= \bar{K}^{\alpha-1} N^{1-\alpha} \gamma^{-\alpha} + (1-\delta) - \frac{\bar{C}}{\bar{K}} \end{cases}$$

$\Rightarrow$

$$\begin{cases} \frac{\theta}{1-N} &= \frac{\bar{K}}{\bar{C}} (1-\alpha) D \\ 1 &= D \gamma^{-\alpha} + (1-\delta) - \frac{\bar{C}}{\bar{K}} \end{cases}$$

Thus, we have

$$\begin{cases} \frac{\bar{C}}{\bar{K}} &= D \gamma^{-\alpha} - \delta \\ N &= 1 - \frac{\theta}{(1-\alpha)D} (D \gamma^{-\alpha} - \delta) \end{cases}$$

where $D \equiv \frac{1}{\alpha} \left(\frac{\gamma}{\beta} + \delta - 1 \right)$.

N_t keeps constant at the steady state. So we do not need to detrend N_t . In other words, N_t doesn't respond as X_t changes at the steady state. This comes from the form of the utility function. Given the King-Plosser-Rebelo form, the income effect and the substitution effect will be canceled out as X_t changes.

(E) LOG-LINEARIZING.

Denote $a_t \equiv \ln(A_t)$, $n_t \equiv \ln(N_t)$, $c_t \equiv \ln(\bar{C}_t)$, $k_t \equiv \ln(\bar{K}_t)$. Let $\gamma = 1$, we get the detrended system as

$$\frac{1}{\bar{C}_t} = \beta E_t \frac{1}{\bar{C}_{t+1}} (\alpha A_{t+1} \bar{K}_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \quad (10)$$

$$\frac{\theta}{1-N_t} = (1-\alpha) \frac{1}{\bar{C}_t} A_t \bar{K}_{t-1}^{\alpha} N_t^{-\alpha} \quad (11)$$

$$\bar{K}_t = A_t \bar{K}_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta) \bar{K}_{t-1} - \bar{C}_t \quad (12)$$

For equation (10), let's define

$$F(\bar{C}_t, \bar{C}_{t+1}, \bar{K}_t, N_{t+1}, A_{t+1}) \equiv \beta \frac{\bar{C}_t}{\bar{C}_{t+1}} (\alpha A_{t+1} \bar{K}_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \quad (13)$$

Take the first order derivatives, and take the value at steady state, we have²

$$\begin{cases} F^{ss} = 1 \\ F_1^{ss} = \frac{\beta}{\gamma} \frac{1}{\bar{C}_{t+1}} (\alpha A_{t+1} \bar{K}_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta)|_{ss} = \frac{1}{\bar{C}} \\ F_2^{ss} = -\beta \frac{\bar{C}_t}{\bar{C}_{t+1}^2} (\alpha A_{t+1} \bar{K}_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta)|_{ss} = -\frac{1}{\bar{C}} \\ F_3^{ss} = \beta \frac{\bar{C}_t}{\bar{C}_{t+1}} \alpha (\alpha - 1) A_{t+1} \bar{K}_t^{\alpha-1} N_{t+1}^{1-\alpha} \bar{K}_t^{-1}|_{ss} = \beta \alpha (\alpha - 1) D \frac{1}{\bar{K}} \\ F_4^{ss} = \beta \frac{\bar{C}_t}{\bar{C}_{t+1}} \alpha (1 - \alpha) A_{t+1} \bar{K}_t^{\alpha-1} N_{t+1}^{1-\alpha} N_t^{-1}|_{ss} = \beta \alpha (1 - \alpha) D \frac{1}{N} \\ F_5^{ss} = \beta \frac{\bar{C}_t}{\bar{C}_{t+1}} \alpha \bar{K}_t^{\alpha-1} N_{t+1}^{1-\alpha} K_t^{-1}|_{ss} = \beta \alpha D \frac{1}{A} \end{cases}$$

²Notice that at steady state, $A_t = 1$.

where $D \equiv \frac{1}{\alpha} \left(\frac{1}{\beta} + \delta - 1 \right)$.

Thus, we have the log-linearized result of equation (10) as

$$0 = \hat{c}_t - \hat{c}_{t-1} + (\alpha - 1)(1 - \beta(1 - \delta))\hat{k}_t + (1 - \alpha)(1 - \beta(1 - \delta))\hat{n}_{t+1} + (1 - \beta(1 - \delta))\hat{a}_{t+1} \quad (14)$$

For equation (11), let's define

$$G(\bar{C}_t, \bar{K}_{t-1}, N_t, A_t) \equiv \frac{(1 - \alpha)}{\theta} \frac{1 - N_t}{\bar{C}_t} A_t \bar{K}_{t-1}^\alpha N_t^{-\alpha} \quad (15)$$

Take the first order derivatives,

$$\begin{cases} G^{ss} = 1 \\ G_1^{ss} = -\frac{(1-\alpha)}{\theta} \frac{1-N_t}{\bar{C}_t^2} A_t \bar{K}_{t-1}^\alpha N_t^{-\alpha} |_{ss} = -\frac{1}{\bar{C}} \\ G_2^{ss} = \alpha \frac{(1-\alpha)}{\theta} \frac{1-N_t}{\bar{C}_t} A_t \bar{K}_{t-1}^{\alpha-1} N_t^{-\alpha} |_{ss} = \frac{\alpha}{\bar{K}} \\ G_3^{ss} = \frac{(1-\alpha)}{\theta} \frac{1-N_t}{\bar{C}_t} A_t \bar{K}_{t-1}^\alpha N_t^{-\alpha} |_{ss} = -\left(\frac{1}{1-N} + \frac{\alpha}{N} \right) \\ G_4^{ss} = \frac{(1-\alpha)}{\theta} \frac{1-N_t}{\bar{C}_t} \bar{K}_{t-1}^\alpha N_t^{-\alpha} |_{ss} = \frac{1}{A} \end{cases}$$

Thus, the log-linearized result of equation (11) is

$$0 = -\hat{c}_t + \alpha \hat{k}_{t-1} - \left(\frac{N}{1-N} + \alpha \right) \hat{n}_t + \hat{a}_t \quad (16)$$

where $N = 1 - \frac{\theta}{(1-\alpha)D} (D - \delta)$ and $D \equiv \frac{1}{\alpha} \left(\frac{1}{\beta} + \delta - 1 \right)$.

For equation (12), we get

$$H(\bar{C}_t, \bar{K}_t, \bar{K}_{t-1}, N_t, A_t) \equiv A_t \bar{K}_{t-1}^\alpha N_t^{1-\alpha} + (1 - \delta) \bar{K}_{t-1} - \bar{C}_t - \bar{K}_t \quad (17)$$

Take the first order derivatives,

$$\begin{cases} H^s_s = 0 \\ H_1^{ss} = -1 \\ H_2^{ss} = -1 \\ H_3^{ss} = \alpha A_t \bar{K}_{t-1}^{\alpha-1} N_t^{1-\alpha} + (1 - \delta) |_{ss} = \alpha D + (1 - \delta) = \frac{1}{\beta} \\ H_4^{ss} = (1 - \alpha) \bar{K}_{t-1}^\alpha N_t^{-\alpha} |_{ss} = (1 - \alpha) D \frac{\bar{K}}{N} \\ H_5^{ss} = \bar{K}_{t-1}^\alpha N_t^{1-\alpha} |_{ss} = \bar{K} D \end{cases} \quad (18)$$

So we have the log-linearized equation as

$$0 = -\bar{C} \hat{c}_t - \bar{K} \hat{k}_t + \frac{1}{\beta} \bar{K} \hat{k}_{t-1} + (1 - \alpha) D \bar{K} \hat{n}_t + \bar{K} D \hat{a}_t$$

$\Rightarrow$

$$0 = -(D - \delta)\hat{c}_t - \hat{k}_t + \frac{1}{\beta}\hat{k}_{t-1} + (1 - \alpha)D\hat{n}_t + D\hat{a}_t$$

where $D \equiv \frac{1}{\alpha} \left(\frac{1}{\beta} + \delta - 1 \right)$.

(F) IMPULSE ANALYSIS.

The differences come from the rational expectation effects. When the agents observe the shock ε_1 , the positive shock also changes the level of A_t . Thus, output, consumption and other variables will increase. But, when the agents observe the shock η_1 , they realize that they will be richer in the future while the technology today still the same. So they increase their consumption and leisure, which will reduce the labor input, output, and capital accumulation, just as that shown in the graph.